

HOPE-RESTORED INVENTORY TRACKING

HOPE RESTORED MISSIONS

SPRING 2025

TEAM MEMBERS & MAJORS

Joshua Koshy: Computer Science
Sanvi Gudipati: Computer Science
Vishnu Jegan Babu: Computer Science
Maheera Mufti: Software Engineering
Divyanshu Krishna Shivran: Computer Science

PROJECT DESCRIPTION

Hope Restored Missions receives donations from local businesses, churches, and individuals. Currently, they are tracking their inventory manually, but it is not as efficient as an online tracking system. In addition to making operations more difficult, this is an issue for soliciting donations and grants (as it is difficult to show how effective they are when they cannot report the exact number of hygiene and food kits given out per year or quarter, for example).

We are tasked to work with Hope Restored to develop an inventory tracking software system that suits their needs and workflow and allows them to keep track of the inflow/outflow of items in house.

PARTNER DESCRIPTION

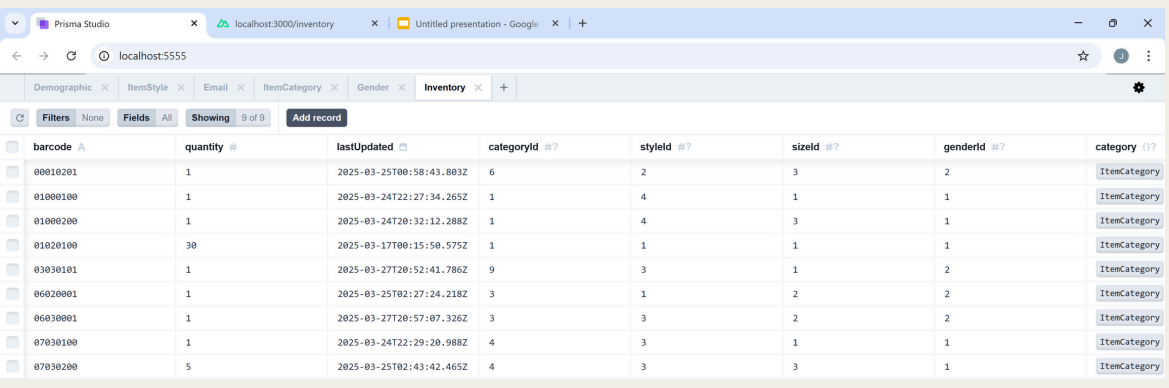
Hope Restored Missions exists to restore hope to those oppressed by the circumstances and tragedies of their lives. We focus on the homeless and those facing housing insecurities. On a foundation of love and support, Hope Restored Missions helps the most vulnerable in our community by meeting their immediate needs, building relationships, and providing resources that put them on a path to self-sustainability.

IMPACT

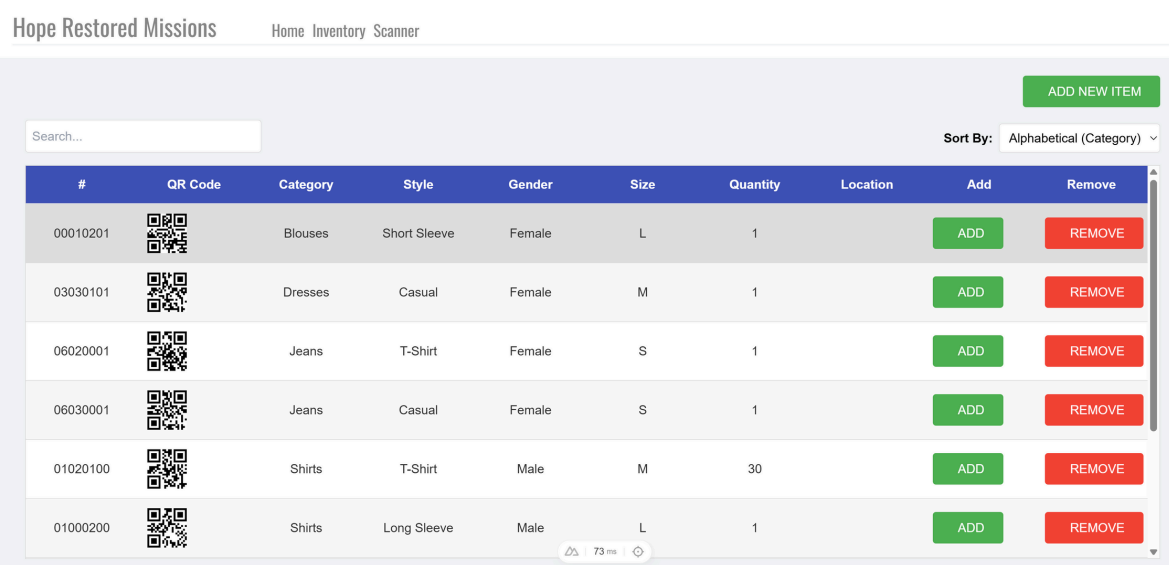
- Real-time stock visibility empowers employees to inform donors and prevent overstocking or shortages.
- Scalable design with database integration supports increased demand and growing inventory.
- Accurate reporting enables easy sharing of inventory insights with stakeholders and donors.
- Bulk item management allows adding or removing multiple items at once, simplifying and speeding up the data entry process.

SEMESTER ACCOMPLISHMENTS

- Set up backend from scratch: configured database, established connection, and integrated live data display.
- Added features to add and remove several items from the inventory.
- Implemented advanced sorting and searching for the database
- Developed features to edit (add or remove from) existing items in the inventory.
- Updated dropdowns in “Add Inventory Item” form to be responsive and reflective of the current inventory.
- Updated inventory category, style, and size options as per sponsor requirements.



barcode	quantity	backordered	category	style	size	gender	location
00000001	1	2025-03-10 10:00:00	4	3	3	3	3
00000002	1	2025-03-10 10:00:00	4	3	3	3	3
00000003	1	2025-03-10 10:00:00	4	3	3	3	3
00000004	1	2025-03-10 10:00:00	4	3	3	3	3
00000005	1	2025-03-10 10:00:00	4	3	3	3	3
00000006	1	2025-03-10 10:00:00	4	3	3	3	3
00000007	1	2025-03-10 10:00:00	4	3	3	3	3
00000008	1	2025-03-10 10:00:00	4	3	3	3	3
00000009	1	2025-03-10 10:00:00	4	3	3	3	3
00000010	1	2025-03-10 10:00:00	4	3	3	3	3
00000011	1	2025-03-10 10:00:00	4	3	3	3	3
00000012	1	2025-03-10 10:00:00	4	3	3	3	3
00000013	1	2025-03-10 10:00:00	4	3	3	3	3
00000014	1	2025-03-10 10:00:00	4	3	3	3	3
00000015	1	2025-03-10 10:00:00	4	3	3	3	3
00000016	1	2025-03-10 10:00:00	4	3	3	3	3
00000017	1	2025-03-10 10:00:00	4	3	3	3	3
00000018	1	2025-03-10 10:00:00	4	3	3	3	3
00000019	1	2025-03-10 10:00:00	4	3	3	3	3
00000020	1	2025-03-10 10:00:00	4	3	3	3	3



#	QR Code	Category	Style	Gender	Size	Quantity	Location	Add	Remove
00010201		Biscuits	Short Sleeve	Female	L	1		<button>ADD</button>	<button>REMOVE</button>
03030101		Dresses	Casual	Female	M	1		<button>ADD</button>	<button>REMOVE</button>
06020301		Jeans	T-Shirt	Female	S	1		<button>ADD</button>	<button>REMOVE</button>
06030301		Jeans	Casual	Female	S	1		<button>ADD</button>	<button>REMOVE</button>
01020100		Shirts	T-Shirt	Male	M	30		<button>ADD</button>	<button>REMOVE</button>
01000200		Shirts	Long Sleeve	Male	L	1		<button>ADD</button>	<button>REMOVE</button>

Remove Inventory Item

Item Barcode: 00010201

Quantity:

Save Cancel

Add Inventory Item

Item Barcode: 00010201

Quantity:

Save Cancel

FUTURE WORK

- Enhanced Category Search
- Fix sort features: newest to oldest, oldest to newest.
- Charts that reflect trends from the database.
- Add additional categories such as food and hygenics if required.
- Potentially implement QR code scanning and printing feature (if project expands).

